

Nuclear Fuel Performance

NE-533
Spring 2026

Last Time

- Problem session for Exam 2
- Started on radiation effects in UO₂
- Main defect type is anion Frenkel pairs
- Charge matters in defects in UO₂! Need local charge neutrality
- Form dislocation loops and voids from radiation damage
- Radiation effects both chemistry and physical properties
- Thermal conductivity degradation with burnup + empirical models

UO2 Thermal Conductivity

- One empirical model used in BISON is the NFIR model
- The model is a function of the temperature T (in °C) and the burnup β (in MWD/kgU)

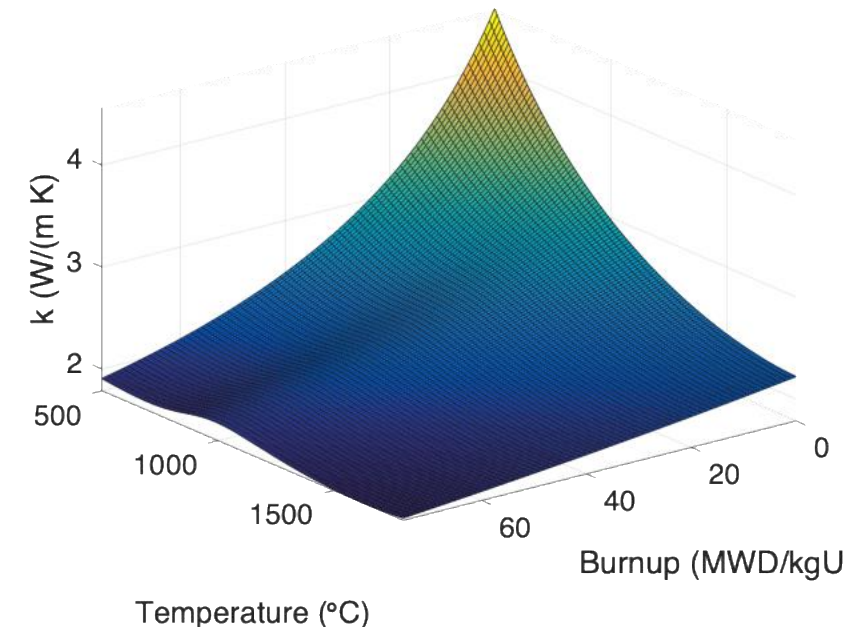
$$k = (1 - R_f(T))k_{ph1}(T, \beta) + R_f(T)k_{ph2}(T, \beta) + k_{el}(T)$$

$$R_f(T) = \frac{1}{2} \left(1 + \tanh \left(\frac{T - 900}{150} \right) \right)$$

$$k_{ph1} = \frac{1}{(9.592 \times 10^{-2} + 6.14 \times 10^{-3}\beta - 1.4 \times 10^{-5}\beta^2 + (2.5 \times 10^{-4} - 1.81 \times 10^{-6}\beta)T)}$$

$$k_{ph2} = \frac{1}{(9.592 \times 10^{-2} + 2.6 \times 10^{-3} \cdot \beta + (2.5 \times 10^{-4} - 2.7 \times 10^{-7}\beta)T)}$$

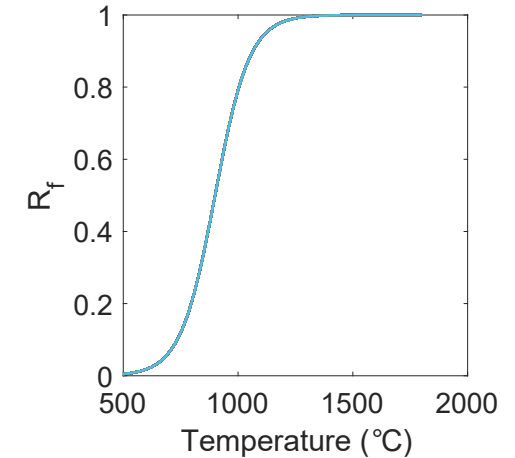
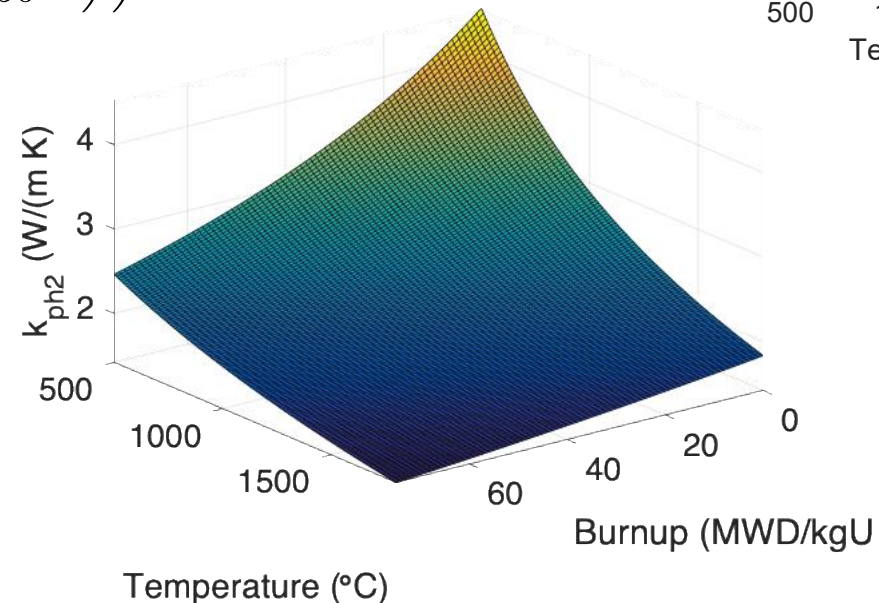
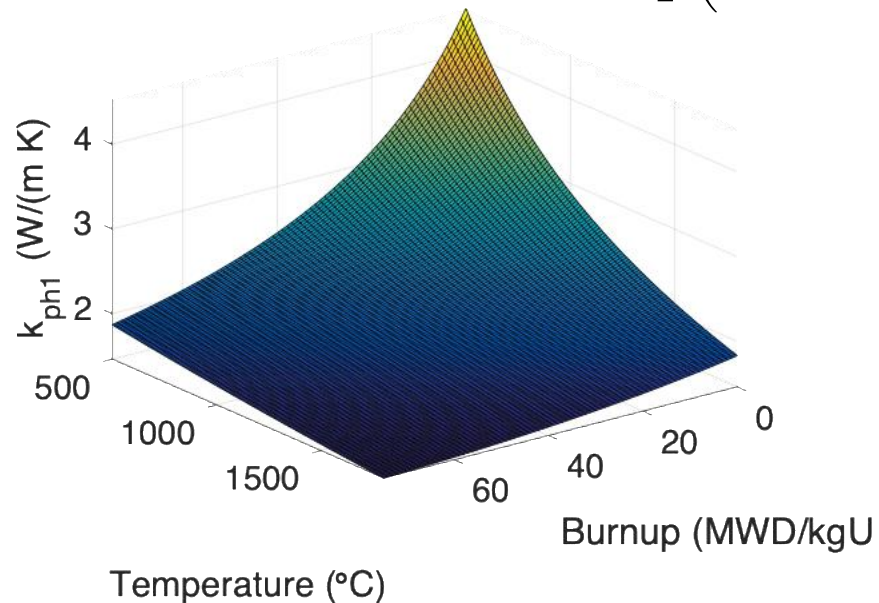
$$k_{el} = 1.32 \times 10^{-2} e^{1.88 \times 10^{-3} T}$$



UO2 Thermal Conductivity

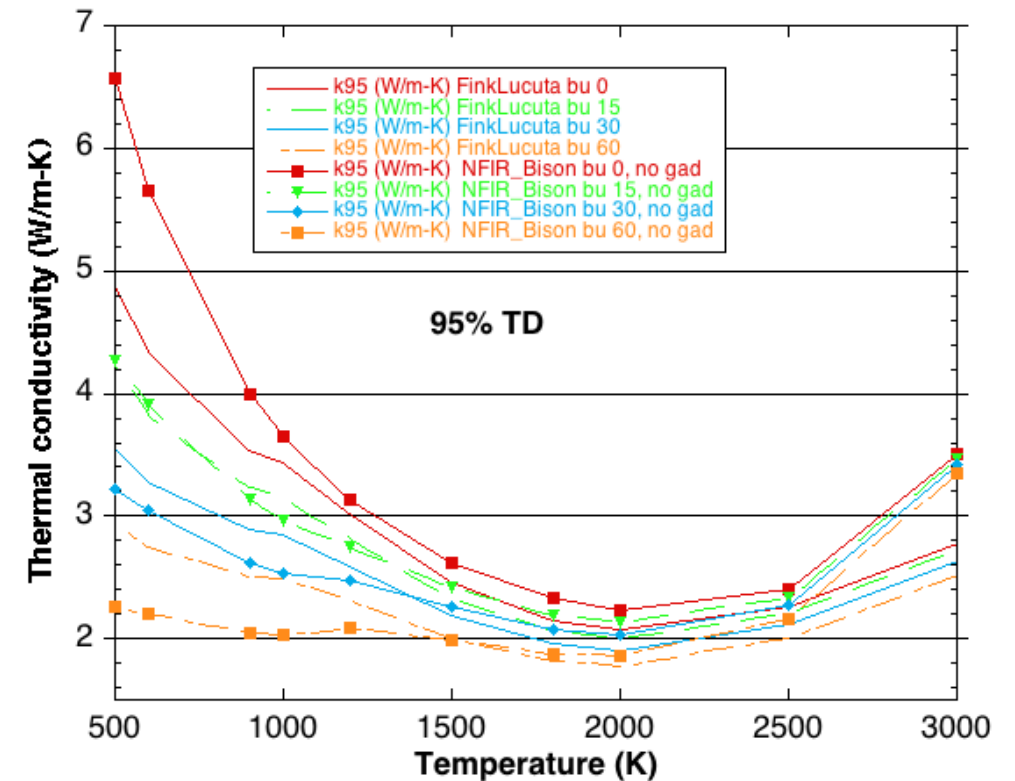
- The R_f function switches between two k_{ph} functions, accounting for thermal recovery

$$R_f(T) = \frac{1}{2} \left(1 + \tanh \left(\frac{T - 900}{150} \right) \right)$$



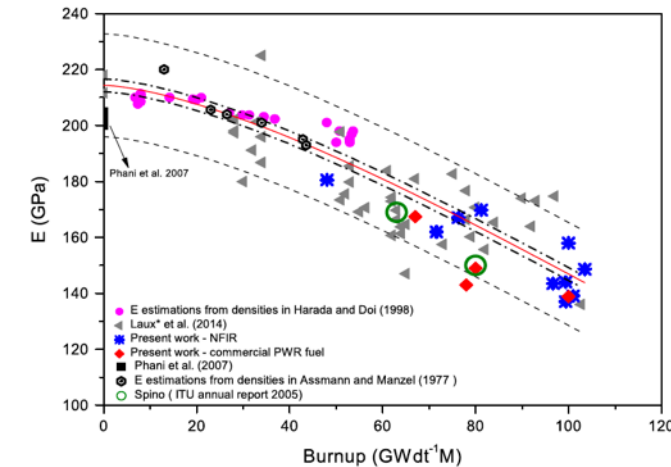
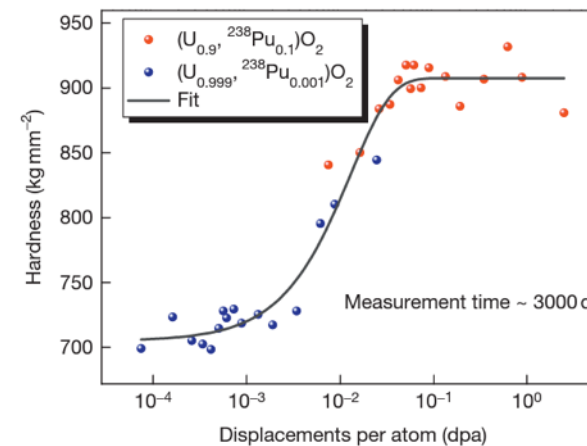
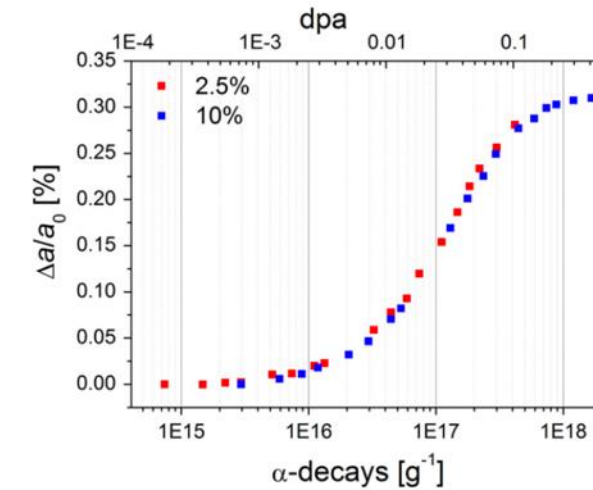
MOOSE/BISON Models

- <https://mooseframework.inl.gov/bison/source/materials/UO2Thermal.html>
- Fink-Lucuta
- Halden
- NFIR
- NFI
- Ronchi-Staicu
- Toptan (NCSU alum)



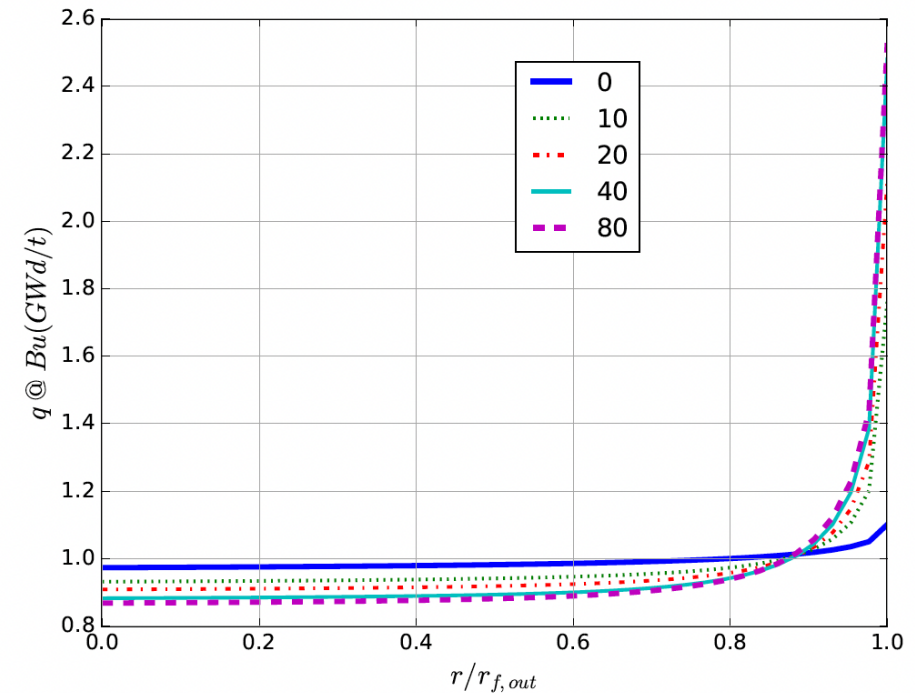
Other Physical Changes

- An important effect of radiation damage is the volume increase of the crystal lattice, leading to macroscopic swelling of the fuel material
- Alpha particle damage alone can cause approximately 0.4% swelling
- Mechanical properties also vary with damage
- Hardness increases as a function of dpa
- Young's modulus decreases with dpa



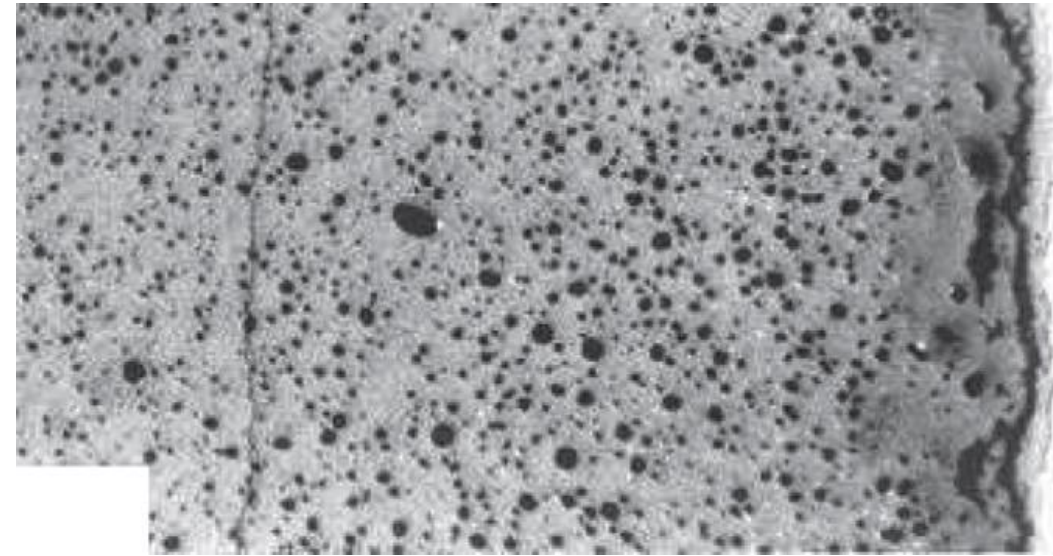
High Power/Burnup on Periphery

- At the beginning of the irradiation, the local power is ~10% higher on the pellet periphery, due to self-shielding effects driven by neutrons entering the rods after interactions in the water
- With increasing burnup, a strong capture of neutrons by U-238 in the resonance range occurs at the periphery of the fuel
- This leads to the production of Pu-239, and leads to a higher fissile density and thus a further increase in the local power/burnup
- This high local burnup area is on the outer ring of the fuel and is about 200 microns thick, which represents about 8% of the fuel



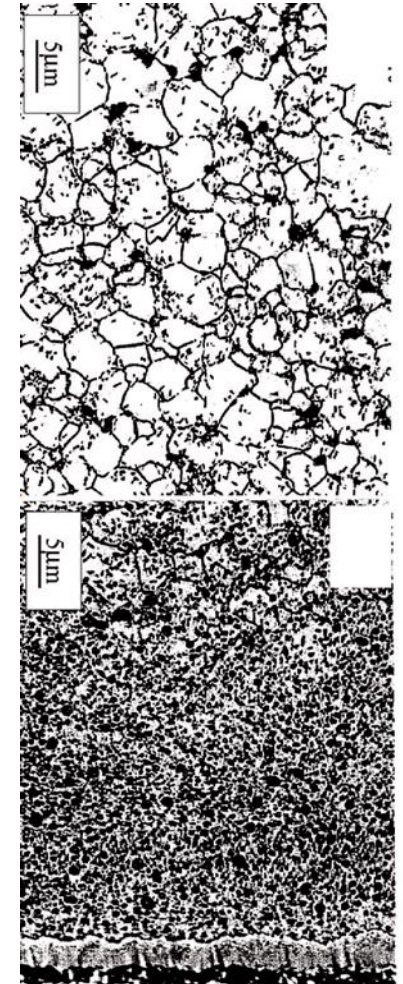
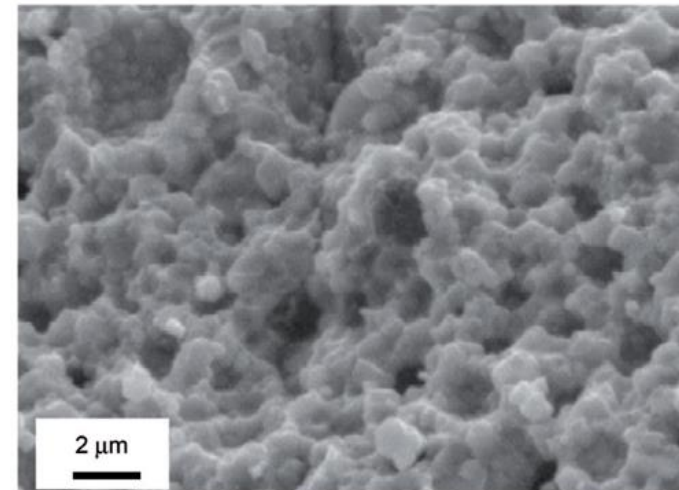
High Burnup Structure (HBS)

- When thermal recovery is not sufficiently efficient (at lower temperatures), the accumulation of defects at high burnup can lead to the instability of the crystalline structure, initiating a restructuring driven by the energy stored in the material
- This can be amorphization (not observed in UO_2) or recrystallization/polygonization
- Polygonization is the rearrangement of dislocations into sub-domain walls with slightly misaligned grains
- In UO_2 , grains subdivide from 10 microns in size to 100-200 nm size, and a densely porous structure is formed (~20% porosity)



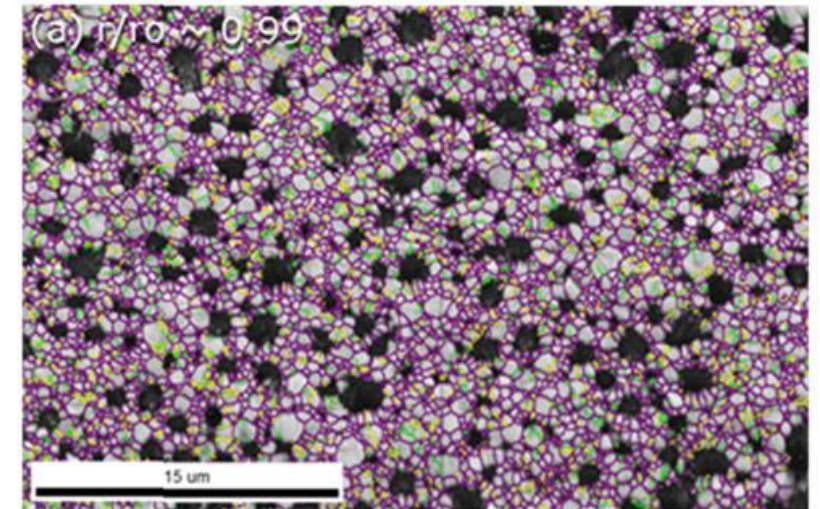
High Burnup Structure (HBS)

- The increase of the relative porosity volume and reduction in grain size degrades the material conductivity
- On the other hand, the intragranular irradiation defects are cleaned, which improves the fuel intrinsic thermal conductivity
- Fission gas in HBS bubble is retained, not released
- Exact mechanism of HBS formation is not fully understood, believe to be polygonization, interacting with fission gas bubbles, dislocation growth, and high levels of energy deposition



High Burnup Structure (HBS)

- HBS has a high fission gas retention capacity, and does not evolve toward an open system of interconnected channels, even when porosity reaches very high values
- Thus, the HBS is a relatively stable structure, limiting pressure increase in the plenum
- While burnup is higher in the rim region, the HBS polygonization removes/sweeps defects to new grain boundaries, removing some phonon scatterers
- Despite high porosity, thermal conductivity slightly increases, slightly decreasing centerline temperatures in high burnup fuel



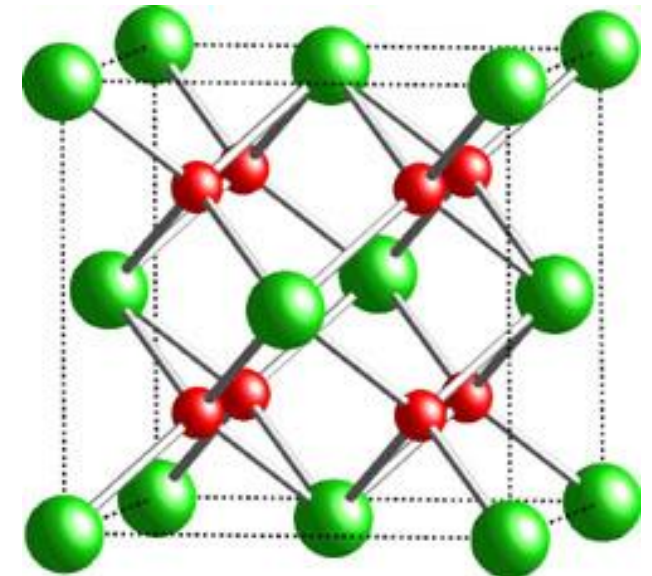
2 - 5° LAGBs shown in green,
5 - 15° LAGBs in yellow,
15 - 65° LAGBs in purple

	Min	Max
15 - 65°	15°	65°
5 - 15°	5°	15°
2 - 5°	2°	5°

FUEL CHEMISTRY

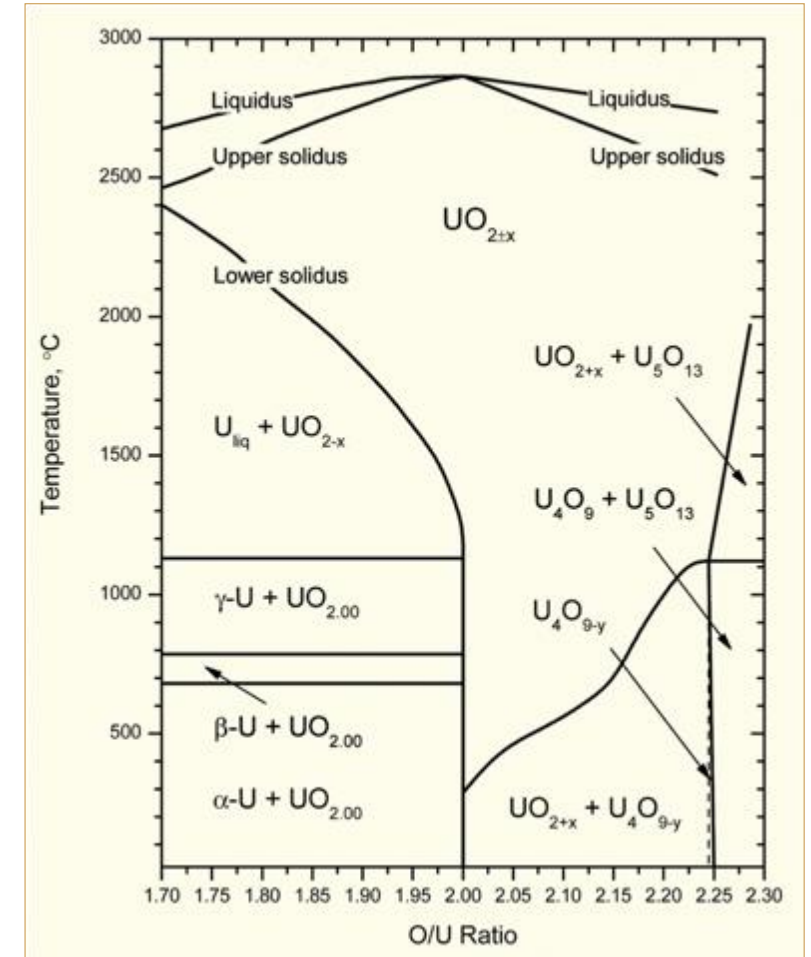
Fuel Chemistry

- UO_2 is an ionic compound that must have balanced charges
- What is the charge of a typical oxygen ion?
 - O^{2-}
- Uranium valence states
 - Possible: U^{3+} , U^{4+} , U^{5+} , U^{6+}
 - Most stable: U^{4+} , U^{5+} , U^{6+}
 - Beyond UO_2 , can have U_4O_9 , U_3O_8 , UO_3
- The UO_2 structure is very stable all the way up to the melting temperature and down to extremely low temperatures, even with irradiation damage
- There is space in the uranium lattice that can accommodate fission products



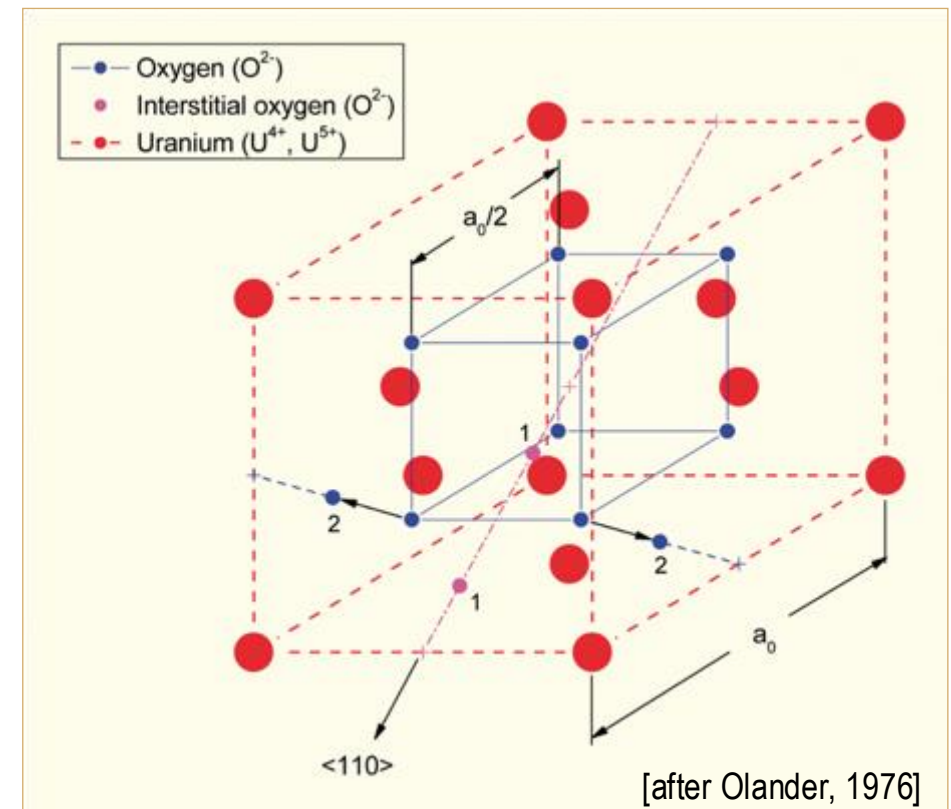
O/M ratio

- The ratio of oxygen to uranium metal (O/M ratio) can vary
- This is the stoichiometry
- The homogeneity range of uranium dioxide extends to both hypo-(UO_{2-x}) and hyper(UO_{2+x})-stoichiometric compositions in oxygen
- The minimum and maximum oxygen contents in the dioxide correspond to the compounds with the formula of respectively $\text{UO}_{1.67}$ at 2720 K and $\text{UO}_{2.25}$ at approximately 2030 K
- Will the O/M ratio go up or down during reactor operation?
 - It is complicated, because of the formation of fission products that also react with the oxygen, and O interaction with Zr cladding



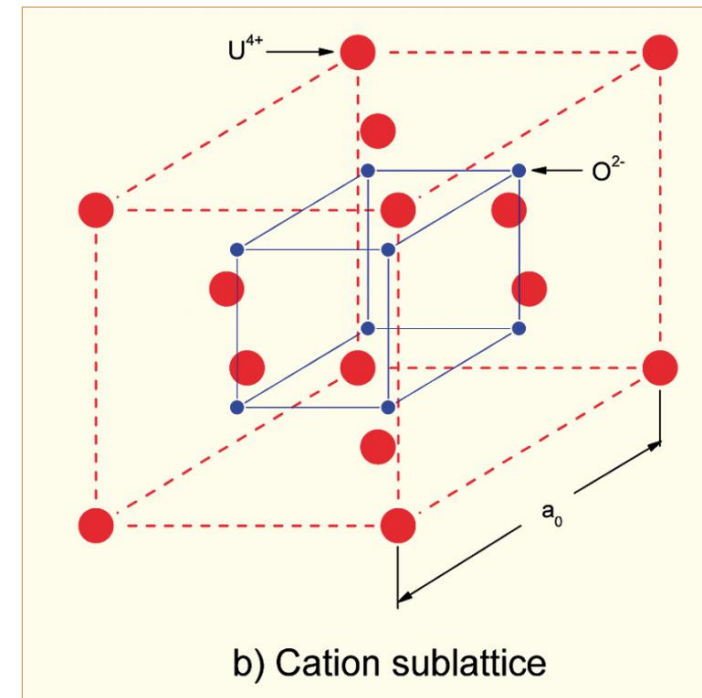
Excess O

- Fission is releasing two oxygen atoms
- The crystal structure of UO_2 can accommodate extra oxygen
- Excess oxygen resides at interstitial locations
- Oxygen in neighboring sites is displaced
- Cation valence increases to maintain electrical neutrality
- Fuel fabricated to be nearly stoichiometric; i.e., $\text{UO}_{2.00 \pm}$ because:
 - It is the most stable
 - It has the highest melting temperature



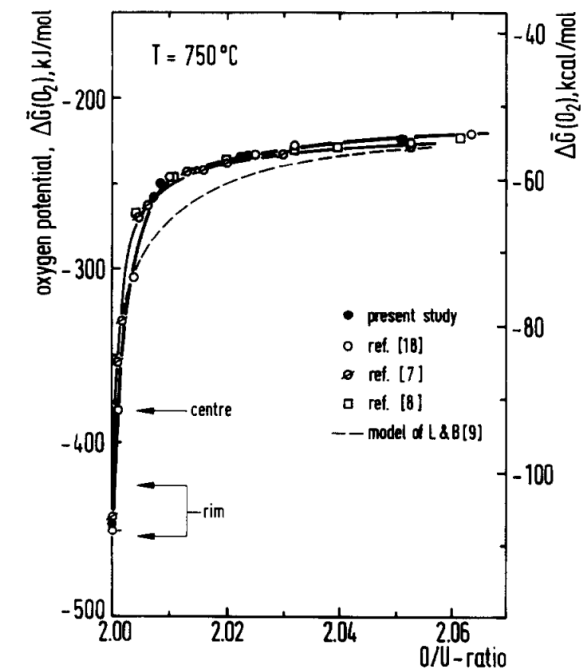
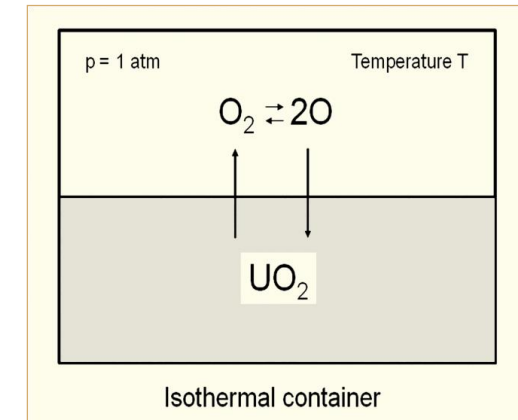
Incorporation of Fission Products

- As fission products form, the valence state of the uranium can change
- Typical valence of soluble fission products is M^{3+}
- The uranium valence state changes to compensate
 - Oxygen liberated by fission
 - Fission products produced with M^{3+} valence state incorporated in fuel lattice
 - Uranium oxidizes from U^{4+} to U^{5+} or U^{6+} to maintain local electrical neutrality



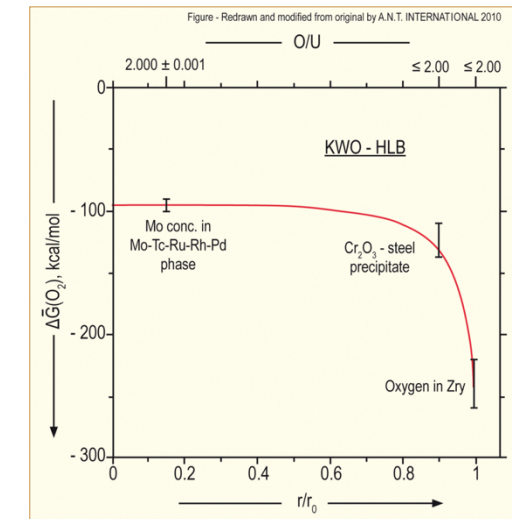
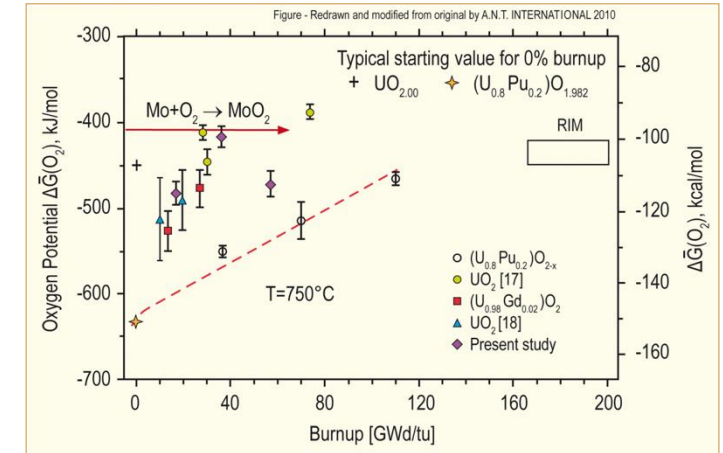
Oxygen potential

- The oxygen potential is a measure of how free the oxygen is to escape/move around
- Oxygen potential is defined as: $\Delta \bar{G}_{O_2} \approx RT \ln(p_{O_2})$
- or, equivalently: $\Delta \bar{G}_{O_2} = 2\mu_{O_{solution}} - G_{O_2}^\circ$
 - With $\mu_{O_{solution}}$ = Chemical potential of oxygen in solution
 - $G_{O_2}^\circ$ = Gibbs free energy of gaseous oxygen at temperature T and a standard pressure (1 atm)
- It is possible to determine the stoichiometry from the oxygen potential



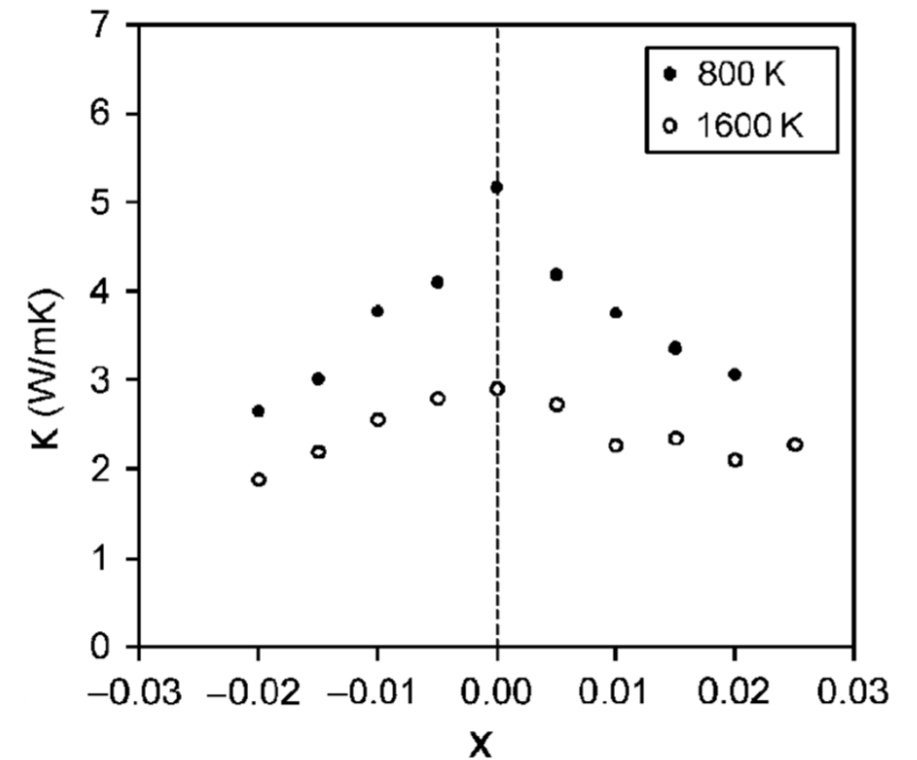
Oxygen potential

- The oxygen potential changes during irradiation, indicating change in the O/M ratio
- Oxygen potential changes during irradiation due to
 - Liberation of oxygen by fission; Generation of fission products; Conversion of uranium to plutonium; Reaction of oxygen with U, Pu, fission products, and cladding
- Oxygen potential across pellet radius observed to be constant at the approximate value of Mo/MoO₂ reaction (from calculations)
- Mo serves as a buffer to the O potential, or a means of inferring what the oxygen potential may have been in the fuel from PIE
- Oxygen potential is low near the cladding, because the oxygen enters the cladding



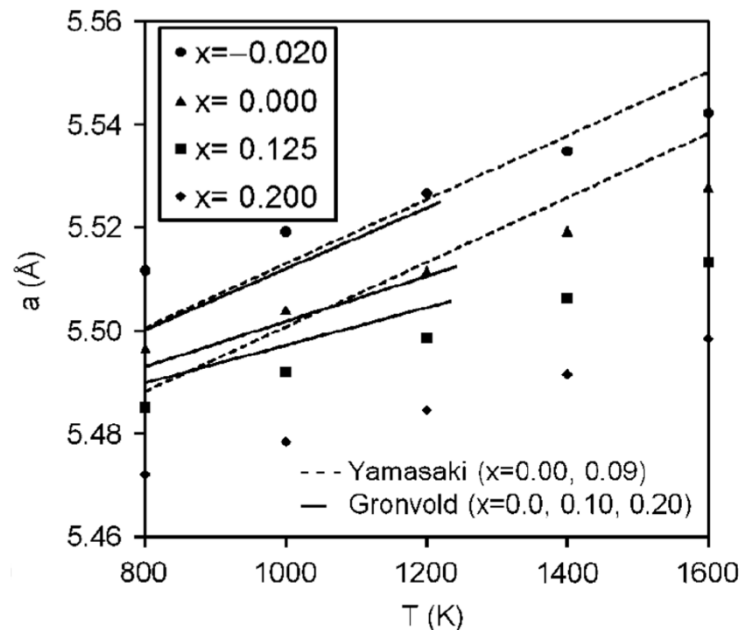
Fuel Stoichiometry/Properties

- The stoichiometry of the fuel directly impacts the fuel performance
- Stoichiometry impacts
 - Melting temperature
 - Thermal conductivity
 - Processes dependent on diffusion
 - Grain growth, Fission gas release, Creep
 - Chemical state and behavior of fission products
 - Chemical reactions at inner cladding surface
- Thermal conductivity is highest for stoichiometric UO_2

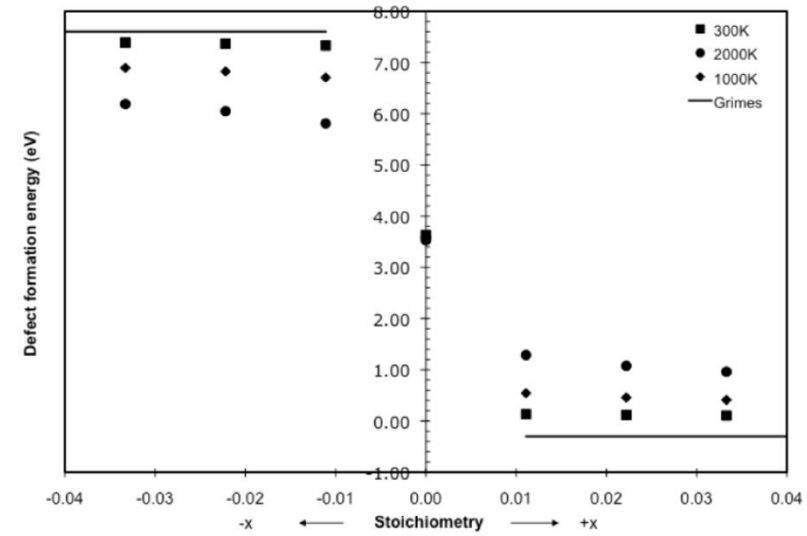


Fuel Stoichiometry/Properties

- The lattice constant of the material decreases with increasing stoichiometry



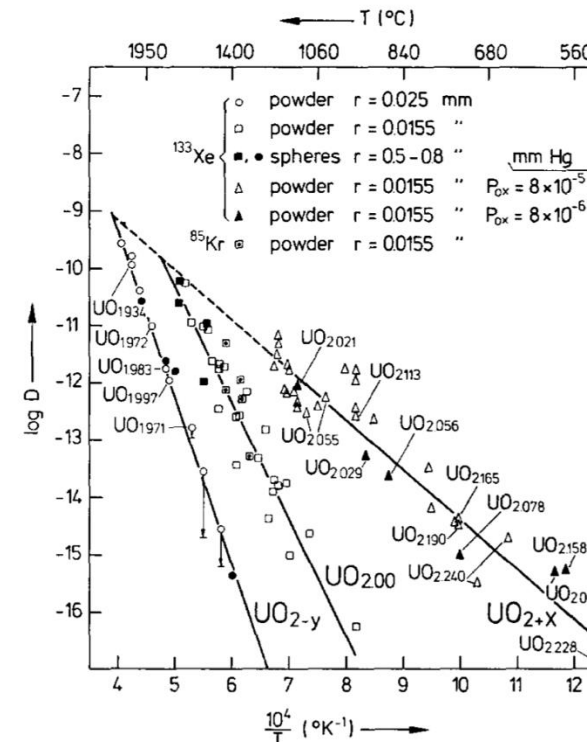
- The uranium vacancy formation energy also changes with stoichiometry



Fuel Stoichiometry/Properties

- The solution energy of Xe, Cs, and Sr in UO_2 depends on stoichiometry as well
- The coefficient defining Xe diffusion also changes with stoichiometry
- Though stoichiometry matters, most fuel performance codes ignore it**

Fission product	$\text{UO}_{1.97}$	UO_2	$\text{UO}_{2.03}$
Xe	3.88 eV	3.88 eV	2.61 eV
Cs	1.7 eV	-0.04 eV	-3.29 eV
Sr	-3.71 eV	-6.03 eV	-9.55 eV



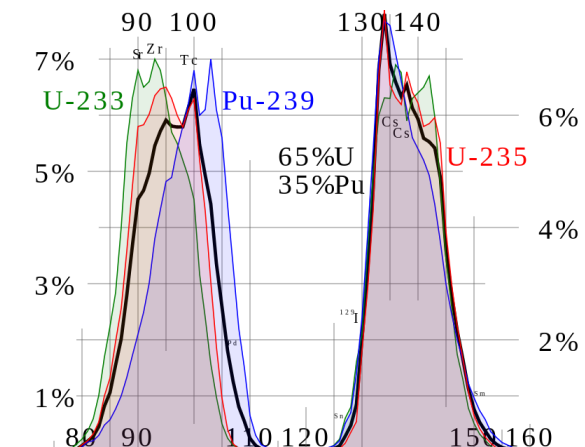
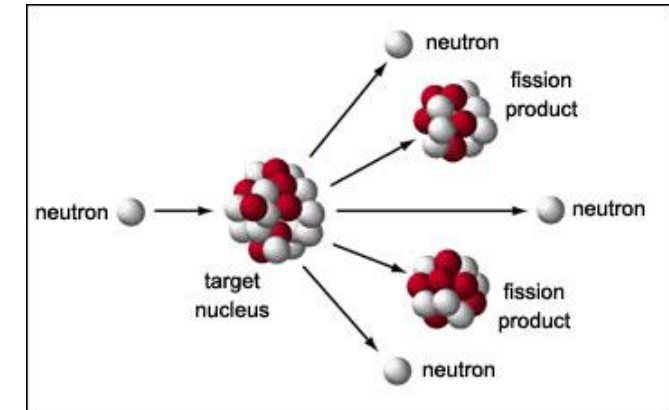
Fuel Chemistry Summary

- UO_2 has a cubic fluorite structure that is very stable
- The charges are balanced with a U^{4+} valence state
- However, the ratio of oxygen to uranium can change. We call this the stoichiometry and abbreviate it as O/M ratio
- The O/M ratio changes during reactor operation, but it is complicated
- The O/M ratio impacts many properties of the fuel

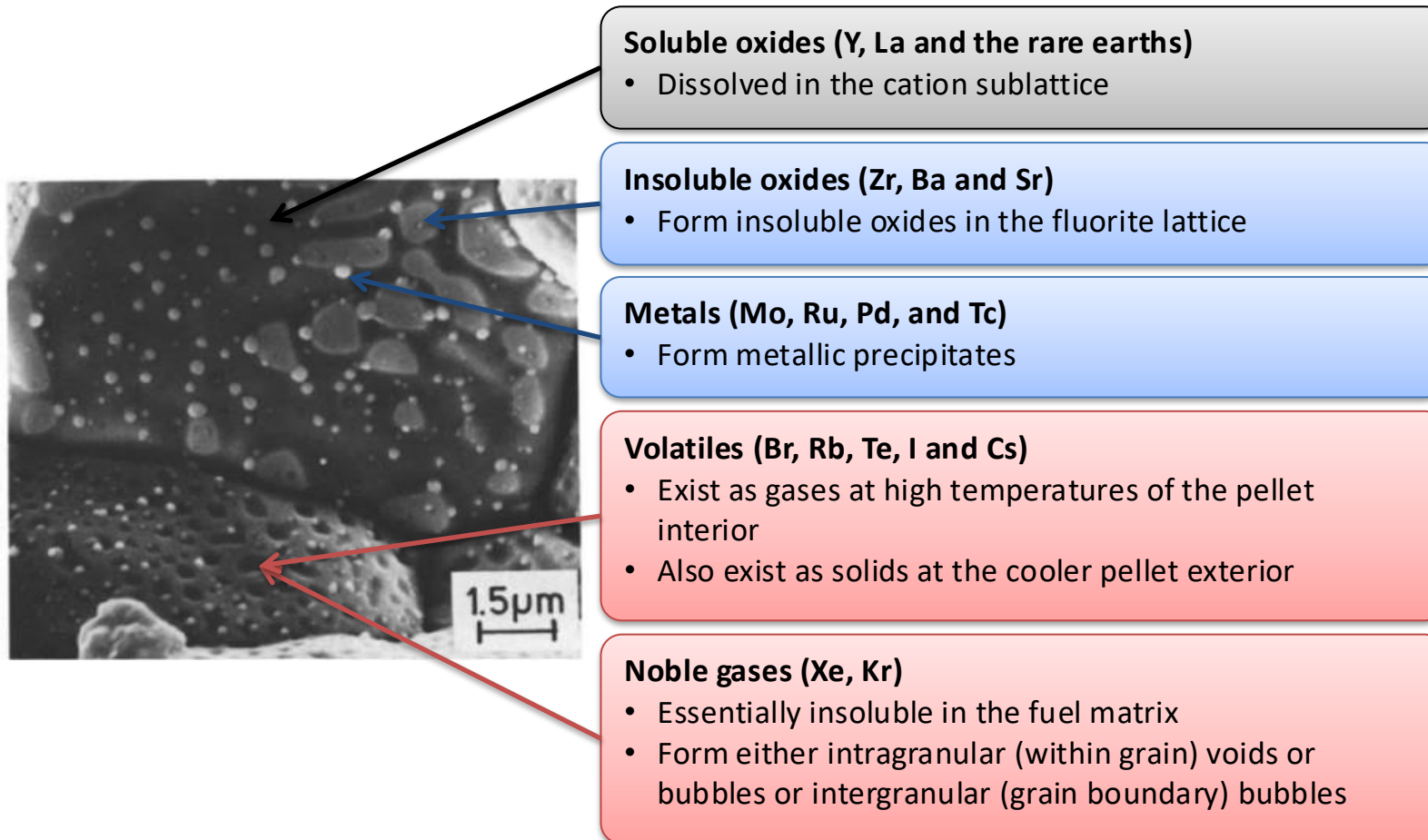
FISSION PRODUCTS

Fission Product Generation

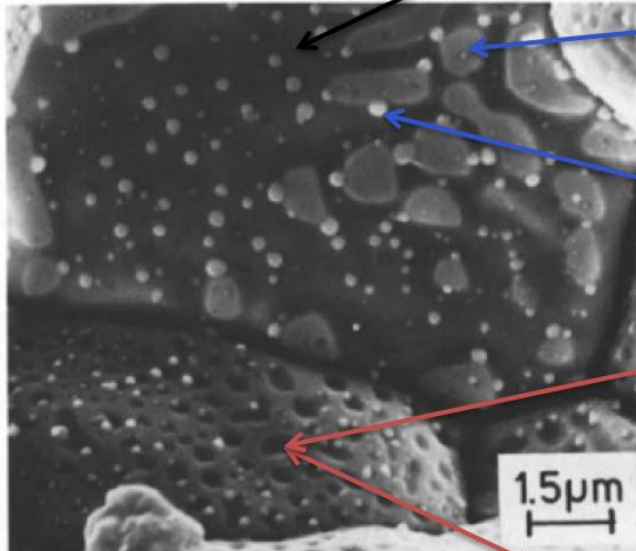
- Fission releases around 200 MeV
 - The fission fragments have 169 MeV of kinetic energy
 - 2 to 3 neutrons with an average energy of 2 MeV
 - 7 MeV of prompt gamma ray photons
 - The remaining energy is released by beta decay
- Every fission product that is produced is now in the crystal lattice of the fuel, changing the chemistry/microstructure



There are various types of fission products that form



All fission products impact the behavior of the fuel

**Soluble oxides (Y, La and the rare earths)**

- Cause swelling, decrease thermal conductivity

Insoluble oxides (Zr, Ba and Sr)

- Can cause swelling

Metals (Mo, Ru, Pd, and Tc)

- Slightly raise thermal conductivity,

Volatiles (Br, Rb, Te, I and Cs)

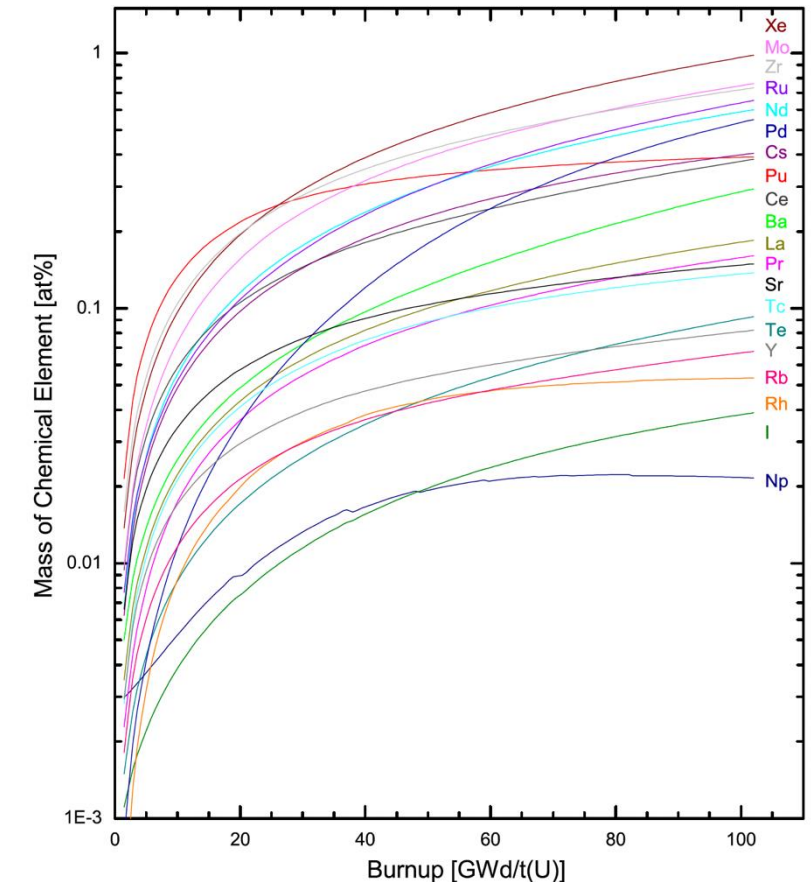
- Cause swelling, decrease thermal conductivity
- Escape from fuel, corrode the cladding

Noble gases (Xe, Kr)

- Cause swelling
- Decrease thermal conductivity
- After release, raise gap pressure and lower thermal conductivity

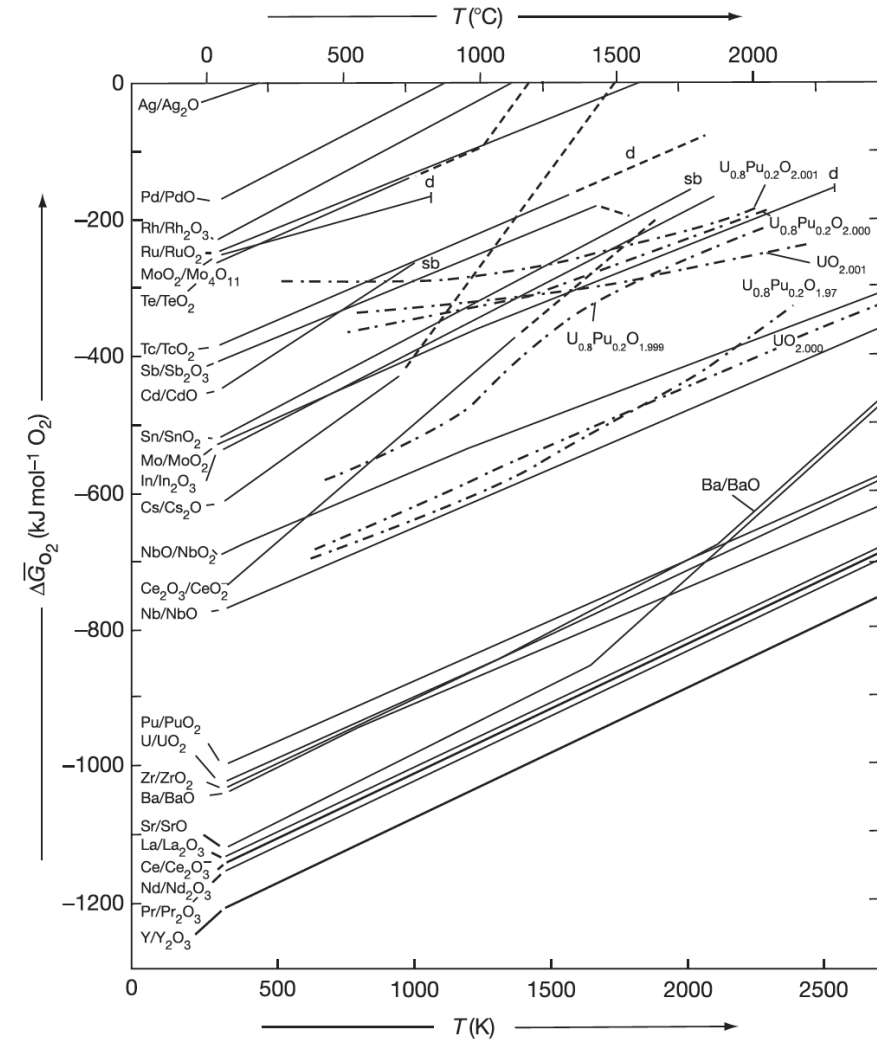
Fission Product Evolution

- Not only are the absolute proportion of fission products accumulating with time, but the relative proportions are changing with respect to one another
- There are continuous transitions between the groups as the critical concentration conditions for new phase development are surpassed because of increased burnup
- Since fission releases two oxygen but only one metal atom, and not all metal atoms will react to bond the oxygen, the O/M ratio of the fuel slowly increases during burnup



Oxide Formation

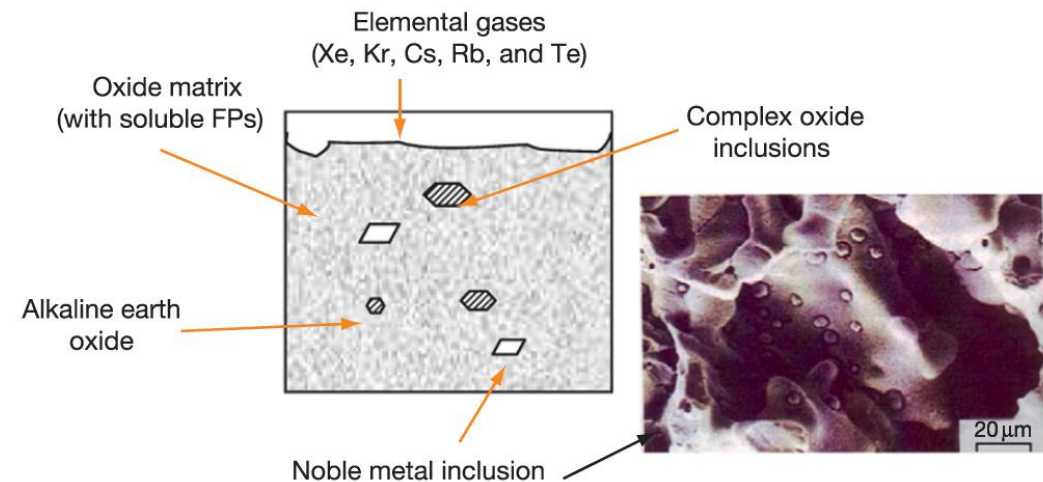
- With burnup, the liberated oxygen will associate with U, Pu, and some Ln but may not be in sufficient supply to combine with Pd and other less oxidizing elements
- If the Gibbs energy of formation of the given fission product lies below the fuel oxygen potential, the element will be capable of forming an oxide (Ce, Sr, Ba)
- If the reaction line is above the oxygen potential in the fuel, the fission product will exist as an element in the fuel (in a separate metallic phase) (Ru, Pd, Tc)



Soluble/Insoluble Fission Products

- When a 4+ valence fission product (e.g., Zr^{4+}) enters the lattice, there is no change in the electrical neutrality
- If the charge of the fission-product cation is lower than U^{+4} , the site occupancy of the lattice is altered to achieve electrical neutrality
- The alkaline earth cations Ba^{+} and Sr^{2+} have large ionic radii and form a separate oxide phase
- Fission products that have limited solubility in UO_2 will segregate to the grain boundaries and voids

Chemical group	Physical state	Probable valence
Zr and Nb	Oxide in fuel matrix; some Zr in alkaline earth oxide phase	4 +
Rare earths	Oxide in fuel matrix	3 +
Ba and Sr	Alkaline earth oxide phase	2 +
Mo	Oxide in fuel matrix or element in metallic inclusion	4 + or 0
Ru, Tc, Rh, and Pd	Elements in metallic inclusion	0 +
Cs and Rb	Elemental vapor or separate oxide phase in cool regions of fuel	1 + or 0
I and Te	Elemental vapor; I may be combined with Cs and CsI	0 or 1 -
Xe and Kr	Elemental gas	0



Fission Product Segregation/Precipitation

- Fission products can be segregated because of their migration to specific locations such as intragranular segregations, grain boundaries, or pellet surfaces
- Metallic inclusions are commonly observed in ceramographs from irradiated samples
- They are formed by isotopes of Mo, Tc, Ru, Rh, Pd, Ag, Cd, In, Sn, Sb, and Te
- These inclusions are found at the surface of the grain boundaries and are associated, in general, with grain boundaries and intragranular bubbles
- Some of the components of these precipitates can be oxidized or reduced forming other compounds that can be released from the fuel matrix
- Cs, Ru, Te, and Ba have been consistently found at the cracks in the pellet and on the clad inner surface
- The white inclusions are metallic precipitates, which are composed of Mo, Tc, Rh, Ru, and Pd, form a quinary alloy in an hcp structure
- A “gray oxide phase” perovskite structure containing different combinations of Ba, Cs, Zr, Mo, and U can also form

Volatile/Gaseous Fission Products

- Xe release occurs at the central region of the pellet where the highest temperatures were achieved during irradiation
- Volatile FPs are released from the fuel matrix similar to that of the noble gases
- Volatile fission products in the gap can react among themselves, resulting in a changing chemical speciation
- The kinetics of formation/decomposition of CsI and Zr iodides are possible factors in the mechanism of SCC

		← A → ← T → ← B →															
I A	II A											III B	IV B	V B	VI B	VII B	
Li	Be											B	C	N	O	F	Ne
Na	Mg	III A	IV A	V A	VI A	VII A		VIII		I B	II B	Al	Si	P	S	Cl	Ar
K	Ca	Sc	Ti	V	Cr	Mn	Fe	Co	Ni	Cu	Zn	Ga	Ge	As	Se	Br	Kr
Rb	Sr	Y	Zr	Nb	Mo	Tc	Ru	Rh	Pd	Ag	Cd	In	Sn	Sb	Te	I	Xe
Cs	Ba	La	Hf	Ta	W	Re	Os	Ir	Pt	Au	Hg	Tl	Pb	Bi	Po	At	Rn
Fr	Ra	Ac															
			Ce	Pr	Nd	Pm	Sm	Eu	Gd	Tb	Dy	Ho	Er	Tm	Yb	Lu	
			Th	Pa	U	Np	Pu	Am	Cm	Bk	Cf	Es	Fm	Md	No	Lr	



Volatile fission product



Metallic precipitates (alloys)



Ceramic precipitates (oxides)



Oxides dissolved in the fuel

FP Interaction with Zr Cladding

- Fission fragments will deposit on the inner surface of the cladding
- Some of these species can diffuse into the cladding, while others can attack the cladding thereby initiating cracks that can later progress with the formation of through-wall cracks
- Oxygen will diffuse into the cladding and contribute to its oxidation state
- Measurements have shown deposits of Sr, Cs, Pu, and Am
- The corrosion behavior of zirconium by Te has been reported
- The chemical reaction between some corrosive fission products and the cladding can lead to PCI
- Fission products such as Cs, Cd, or I can attack the cladding inducing crack initiation, which then progress through the cladding by intragranular and transgranular cracking modes